

# LUCAS SIGNORINI

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## EDUCATION

### Carnegie Mellon University

Pittsburgh, PA

B.S. + Integrated M.S. in Electrical and Computer Engineering

B.S. Expected May 2027 | M.S. Expected May 2028

GPA: 3.67 | Dean's List: Spring 2025, Fall 2025, Spring 2026

**Relevant Coursework:** Structure & Design of Digital Systems, Microelectronic Circuits, Fundamentals of Control, Autonomous Robotics I, Logic Design & Verification\*, Distributed Embedded Systems\*, Intelligent Robotic Systems\*

\*in progress, Fall 2026

## SKILLS

**Languages:** C, C++, Python, SystemVerilog

**Embedded & RTOS:** STM32 bare-metal firmware · custom RTOS design (RMS scheduling, priority-ceiling-protocol mutexes, PendSV context switching) · interrupt-driven UART/GPIO/PWM drivers

**Circuit & PCB Design:** Altium Designer · LTSpice · Cadence Virtuoso · PCB Layout & Schematic Capture (prototype-to-production)

**FPGA & Digital Logic:** SystemVerilog (structural design) · Vivado · FSM design · waveform-level debugging

**Robotics & Controls:** ROS2 · PID control with anti-windup · motor/stepper driver integration

## EXPERIENCE

### Onward Robotics - Electrical Engineering Intern, Hardware Team

Pittsburgh, PA | May 2025 – Aug 2025

- Extended the Lumabot's backup battery runtime from about a month to a year by designing, simulating (LTSpice), and laying out (Altium) a custom disconnect PCB, carrying it from breadboard prototype through a production-ready board.
- Independently redesigned the production board's footprint beyond the original brief, cutting its mounting hardware from three screw points to two.
- As part of an assigned intern presentation on ways the company could improve, pitched a fix for the team's fragile, error-prone parts database; after leadership chose to advance the idea, built it into a Google Forms-based tool and delivered a usage case and step-by-step guide for the team's adoption review.
- Root-caused a voltage-clamping circuit failure by characterizing power dissipation across the 60V protection stage under load, then resolved it by re-specifying component values to bring dissipation within rated limits.

## RESEARCH

### CMU CyLab - Undergraduate Researcher

Pittsburgh, PA | June 2026 – Present

- Independently designing a contactless safety sensor for newly legalizing plug-in solar (PIPV) systems as the sole undergraduate researcher under Prof. Anthony Rowe, evaluating energy-harvesting, battery-powered, and active-signal-injection sensing approaches before selecting a current-transformer-based design.
- Built a contactless current sensor that senses current through an electrical breaker by exploiting the breaker's internal construction, paired with a custom signal-conditioning circuit to prepare the signal for downstream digitization.
- Assembling a physical test bench that replicates a real residential electrical panel (code-rated breakers loaded with electric space heaters) to validate the design under realistic conditions.

## PROJECTS

### Custom Real-Time OS & PID Motor Control (STM32) - CMU 18-349

Spring 2026

- Designed and fabricated a custom PCB (Fusion Electronics) interfacing an STM32 Nucleo board with a car kit's drive motor.
- Built, with a partner, a preemptive, priority-based real-time kernel from scratch in C for ARM Cortex-M, implementing Rate Monotonic Scheduling admission control and an Immediate Priority-Ceiling-Protocol mutex system, validated against an automated test suite covering schedulability, concurrency, and thread-lifecycle correctness.
- Deployed the kernel to drive the car kit with a real-time PID motor controller (anti-windup-tuned), closing the loop on interrupt-driven quadrature-encoder feedback, with live speed/PID telemetry on an onboard LCD and a custom Python tool visualizing measured versus target speed.

### Air Hockey Robot - CMU Robotics Club / Build 18

Jan 2025 – Apr 2026

- Helped build an air hockey robot opponent during an intensive one-week Build 18 competition sprint (earning a dedication award), then extended it afterward with a computer-vision tracking pipeline built from an iPhone camera and a laptop, autonomously tracking the puck and driving the return shot.
- Designed and built the camera's PVC housing and wired the Arduino, motor driver, and linear actuator together, contributing to the Arduino-to-motor-driver control code.
- Diagnosed a motor-overheating failure caused by intentionally overclocking the drive motor for faster response time, then designed and added an active cooling fan to sustain performance without sacrificing speed.

## AWARDS

1st Place, Red Robot Competition (Nov 2023) · 1st Place, E-TRASH Robotics Hackathon (Jan 2025)